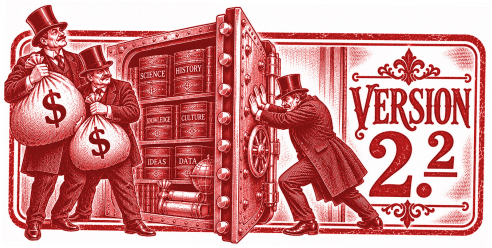
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Fields	Philosophy of AI; social epistemology; science and technology studies; history of information science; AI governance
Keywords	enclosure; knowledge commons; epistemic commons; provenance; contestability; answerability; recognition; objectivity; citation indexing; research assessment; statistical order of knowledge; large language models; evaluative discretion; displacement
Jurisdiction	Not jurisdiction-specific. Legal examples are drawn from the United States and the European Union; the comparative legal audit (United Kingdom; European Union; United States; China) is in the companion paper.
Audience	Researchers in philosophy of AI, social epistemology, science and technology studies, the history of information science and information law; and those who run research assessment, libraries and scholarly communication.
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Abstract

Debates about artificial intelligence (AI) and the knowledge commons have concentrated on the appropriation of training data and on restricted access. This paper argues that a further enclosure concerns the governance of knowledge: who sets the standards by which contributions are selected, justified and recognised, and to whom those standards answer.

Drawing on the history of citation indexing and on social epistemology, the paper separates the origin of a claim, its evidential justification, the justification of the standards that selected it, and the authority that makes it count. Answerability is relative to those affected, and concentration is not enclosure. The third enclosure occurs when unanswerable epistemic intermediation materially displaces a community's effective capacity to sustain and contest the epistemic practices on which it depends: by transfer, when a system's standards come to govern a practice and those it serves cannot stop them, or by suppression, when reliance at scale depletes a valuable, contestably governed public practice into an unanswerable substitute on which they come to depend. Its strict case is the displacement of an epistemic commons. The Science Citation Index shows external metrics entering research evaluation and perhaps, in places, governing it; declining contributions to public question-and-answer forums show a plausible mechanism of suppression. Drawing on Helen Longino, the paper argues that such displacement can be an epistemic loss as well as a political one. AI makes these conditions possible across general-purpose knowledge infrastructure; how far they have produced displacement is an empirical question. A companion paper sets out the remedies.

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About the Author

Peter Kahl is an independent researcher working on the philosophy of AI, law and institutional governance. His research concerns answerability: who can be made to account for the standards by which knowledge is selected, evaluated and recognised. Before turning to law and philosophy, he spent more than two decades in electronics and semiconductor engineering, where every standard answered in the end to the device. He holds an LLB from Staffordshire University and an LLM with Distinction from the University of Reading, where the first version of this paper was his dissertation. He has published peer-reviewed work on agency in AI systems in *Discover Artificial Intelligence*.

About Lex et Ratio

Lex et Ratio is Peter Kahl's independent research and publication platform for work across philosophy, technology, law and institutional governance. It publishes working papers, research software and related research on representation, agency, epistemic governance, AI and answerability.

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WORKING PAPER

The Third Enclosure Movement

PROVENANCE, ANSWERABILITY AND RECOGNITION IN ARTIFICIAL INTELLIGENCE

Peter Kahl, independent researcher, Lex et Ratio
26 October 2026

1. Introduction: ‘Renting back the library’

1.1 Two complaints, sixty years apart

In October 2026 the Barcelona-born historian of knowledge Xavier Nueno won the 54th Anagrama essay prize for *La revuelta de los copistas* (‘The Revolt of the Copyists’), a history of the copyists, stenographers, archivists, librarians and programmers whose work made knowledge storable and findable. Announcing the prize, he is reported to have put the book’s accusation in a single sentence: the technology companies want to rent us back the library itself. The book argues, as reported, that AI firms drew on the collections libraries had preserved, through mass digitisation and the illicit downloading of books, and sold access to the result by subscription; and that, because language models feed on a collective memory, the models should be made public (Gary 2026; Nueno 2026).[1]

The complaint has a partial precedent. In 1980 the Institute for Scientific Information (ISI) moved into new headquarters in West Philadelphia, whose heavily secured computer room held a database of tens of millions of citations extracted from the scientific literature (Nueno 2023, pp 111–112, 137; see also Csiszar 2018). Its product, the Science Citation Index, rested on Eugene Garfield’s idea that authors index their own work through their references: ‘we are in reality utilizing an army of indexers, for every time an author makes a reference he is in effect indexing that work from his point of view’ (Garfield 1955, quoted in Wouters 1999, p 31). The experimental index was supported by the National Institutes of Health and the National Science Foundation before ISI published it commercially from 1964 (Wouters 1999, pp 70, 72–73). The citing scientists were largely publicly funded; the firm that consolidated their references was private; and the index, built as a search tool, was from the late 1960s promoted increasingly as an instrument for evaluating the scientists who had supplied its raw material (Nueno 2023, pp 133–135), its successors’ ranking data being sold to governments and evaluation agencies (Nueno 2023, p 208). When a publisher complained in 1980 that ISI acted as a gatekeeper, the court upheld the firm’s right as a private company to apply its own

[1] The book was published in Spain on 21 October 2026. This paper relies on Nueno’s earlier published work and on reports of the book, and does not characterise the book beyond what those reports state.

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criteria to what it indexed (Nueno 2023, p 138). Nueno calls this ‘a new delimitation of public and private knowledge’ (Nueno 2023, p 114 n 6).

The precedent has a limit that matters. The Science Citation Index preserved provenance more completely than almost any instrument before it: every entry named its source. ISI concentrated authority over the significance of a practice without dissolving its trail. Generative answering is not a repetition of the citation index; it brings together tendencies that indexing, ranking and evaluation had kept apart.

1.2 The question

Version 1 of this paper asked whether AI-driven enclosure required a shift from proprietary regimes towards partial commons, and answered yes (Kahl 2025a). That answer is kept for the enclosures of access. But the ISI precedent shows the question was too narrow. What mattered about ISI was not that it sold access to citations, but that its index made citations countable while presenting the count as involving no human judgement; funders, administrators and universities then turned the count into a measure of quality, and in places, perhaps, into a governing one (section 5.1). The question of this version is therefore:

> *What, exactly, is enclosed when knowledge is consolidated into AI systems, and under what conditions does that consolidation displace the governance of knowledge by those it serves?*

1.3 Thesis

The paper’s principal claim is this: *the third enclosure occurs when unanswerable epistemic intermediation materially displaces a community’s effective capacity to sustain and contest the epistemic practices through which it produces, tests and recognises knowledge*. Contribution remains widely distributed; the authority to select, justify and recognise knowledge becomes concentrated and hard to challenge.

Three operations bear on whether AI intermediation is answerable. The *dissolution of provenance* is a condition that obscures the record; the *disowning of judgement* is a practice that obscures who is responsible, locating standards in a mechanism said to decide nothing; and the *concentration of recognition authority* is a consequence of delegation and reliance. They are enabling features, not elements of the definition: none of them by itself makes a system unanswerable or produces enclosure. What is at risk is two capacities: the *operational* capacity to go on producing, correcting and preserving public contributions, and the *constitutive* capacity to found, sustain and govern the institutions and standards through which contributions are recognised. Enclosure turns on a two-part test (sections 3.6 and 3.7). A system’s standards must be unanswerable to those affected, who can neither share in deciding an inadequately justified standard nor obtain a binding remedy against it through an external route. And the system must displace their capacity, by *transfer*, which removes the capacity to contest a practice’s standards, or by *suppression*, which removes the capacity to sustain it. The strict case is the displacement of an *epistemic commons*, a practice governed by those

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it serves; the wider case includes practices that answered to them without being governed by them. The concept is technology-neutral. What AI changes is architecture and potential reach, and how far the conditions it creates have produced displacement remains an empirical question. The 'third enclosure movement' therefore names a proposed interpretation of an emerging tendency, not a demonstrated historical transformation; section 7.1 sets out what is established at each level.

Why call this enclosure rather than unanswerable displacement? Because of a second-order loss. An unanswerable standard can produce bad decisions. When its adoption also displaces the institutions or capacities through which it could have been resisted, the community loses more than good decisions: it loses its capacity to make, contest or replace the standards that govern it, and the harm tends to reinforce itself, since the means of correcting it are what has been lost. That is what the historical term captures, and why displacement is required: an unanswerable system introduced where no practice existed may be objectionable, but it dispossesses no one. The concept does not ground every duty one might propose; it identifies a distinctive harm and explains why it matters (section 3.8). What follows for governance is the subject of the companion paper (Kahl 2026f).

1.4 What has changed since version 1, and the companion paper

Version 1 was a doctrinal and comparative Master of Laws (LLM) dissertation. This version is philosophy of AI. It draws on the history of automated indexing, chiefly Nueno's account of the 'statistical order of knowledge' (Nueno 2023) and Wouters's history of the citation index (Wouters 1999); on social epistemology, chiefly Longino; and on work on answerability and recognition, including the author's own, from which the inferences this paper needs are made here rather than taken on trust. The sector case studies and the chapter on Rawlsian, Marxist and Foucauldian frameworks have been removed; Ostrom is retained. The comparative legal audit and the remedies are developed in a companion paper, *The Answerable Commons* (Kahl 2026f). This paper establishes the concept; the companion asks what law can see of it and what would answer it.

1.5 Method, scope and plan

The method is conceptual analysis disciplined by history. The scope is systems that rank, retrieve and generate text in answer to questions: search engines, recommender systems and large language models, including those that combine generation with retrieval. Chapter 2 places the third enclosure in its lineage. Chapter 3 separates the questions of origin, justification and authority, sets a relational test of answerability, defines enclosure by transfer or suppression, and says why its object matters. Chapters 4 to 6 take judgement, recognition and the commons in turn. Chapter 7 states what is established and what remains conditional.

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2. Three enclosures

2.1 Land

Between the sixteenth and nineteenth centuries England's common fields were enclosed and customary rights of grazing, gleaning and fuel-gathering extinguished, by local agreement and piecemeal as well as, chiefly between 1750 and 1830, by parliamentary Enclosure Acts (Turner 1986). Karl Polanyi read the process as the first stage of turning land, labour and money into 'fictitious commodities' (Polanyi 2001, pp 71–80); E. P. Thompson showed how it dissolved moral economies of reciprocity that had governed access to the commons (Thompson 1991, ch 3). Law did not merely record a change in who held land; it extinguished particular customary rights and the local institutions through which they were exercised, with effects that varied from place to place. And the change was justified as improvement.

2.2 Information

James Boyle named the 'second enclosure movement': the extension of property rights over informational goods, again justified by the claim that exclusive rights would produce more (Boyle 2003; 2008, ch 3). Its international settlement was the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS); Susan Sell called the result private power made through public law, and Peter Drahos and John Braithwaite 'information feudalism' (Sell 2003; Drahos and Braithwaite 2002). It was resisted by open licensing and by the scholarship on knowledge commons that grew from Ostrom's work (Ostrom 1990; Hess and Ostrom 2007; Frischmann, Madison and Strandburg 2014).

2.3 Between them: the statistical order of knowledge

Accounts of enclosure usually jump from intellectual property to AI. Nueno's history of automated indexing shows what lies between: the attempt, from the turn of the twentieth century, to produce maps of knowledge without the human indexer (Nueno 2023, p 2). Where Dewey's classification divided knowledge into fixed classes, the indexers Nueno studies wanted order to emerge from the documents themselves, above all from citations. He calls the result the 'statistical order of knowledge'. Its promise was that 'the subjectivity of the indexer can be bracketed'; his finding is that this 'was nothing but an illusion', because indexing was delegated to algorithms that 'establish their own ranking criteria' (Nueno 2023, p iv). Bowker and Star had made the general point that classification systems are infrastructures that become invisible in use and, in shaping what institutions can see, distribute advantage (Bowker and Star 1999). ISI then showed a private firm taking over a task previously understood as public or non-profit, and a search technology becoming an evaluation technology (Nueno 2023, pp 113–114). And Google's PageRank was explicitly modelled on citation analysis (Page et al. 1999): 'for the indexer to disappear, it had to become omnipresent', since every user who links, clicks or shares indexes without knowing it (Nueno 2023, p 200).

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2.4 What the third enclosure adds

Nueno ends his thesis by observing that ranked search was being replaced by chatbots that answer with a single generated text, and that the statistical order of knowledge had not ended but 'finally triumphed', constructing information as 'knowledge without a subject' (Nueno 2023, pp 203–204, 211–212). Searching, which once involved 'comparison, selection and judgement', becomes a conversation with a voice that 'appropriates authorship while obscuring its true origins' (Nueno 2025).

This paper defines the third enclosure movement as the material displacement, through AI systems whose standards are unanswerable to those affected, of a community's effective capacity to sustain and contest the epistemic practices on which it depends.[2] The first two enclosure movements concerned rights of access, use and exclusion, and the forms of enclosure of access catalogued in v1 (section 3.1) belong with them. The third concerns the governance of epistemic practices. Governance as such is not new: intellectual property also structured the authority to reproduce, transform and authorise uses of information. What is new is its object, the standards by which epistemic contributions are selected, justified and recognised. Restrictions of access contribute to it only when they produce the displacement section 3.7 specifies: a firm that restricts a valuable dataset encloses access, but encloses governance only if it thereby displaces a community's capacity to sustain or contest a practice. The three operations are contributing conditions of this second kind. The second enclosure extended property over identifiable things; the third takes in a practice, the work of attribution, selection and judgement by which a community makes knowledge findable, trustworthy and open to correction. The ordinal names a proposed historical movement, not the first appearance of the mechanism. The definition is technology-neutral, and earlier arrangements, such as religious censorship, the licensing of printers, colonial systems of classification or monopolies over professional certification, may on this definition have enclosed epistemic practices before. What could make the present case a movement is that AI allows the conditions of enclosure to operate across general-purpose knowledge infrastructure at a scale and degree of standardisation earlier intermediaries lacked (section 7.1); whether they do is the open question. It is not the endpoint of the statistical order of knowledge: retrieval, open-weight release and new attribution techniques show that the configuration is still changing. It may also be recursive, which would matter for the reproduction of public knowledge. Under experimental conditions in which generated data progressively replaced original data, training on recursively generated data degraded models, the rarer parts of the original distribution disappearing first (Shumailov et al. 2024). Whether that is happening is a separate question. The link to suppression is a hypothesis: if public contributions

[2] The phrase is older than its application to AI: Michael Madison used it in 2005, half in jest, for the claiming of snow-cleared parking spaces on public streets, following Boyle's second enclosure (Madison 2005). The author applied it to AI in his Master of Laws (LLM) dissertation, submitted to the University of Reading on 24 August 2025, and in the first version of the present paper, published in November 2025 (Kahl 2025a). Sonia Katyal and Christopher Morten have since presented a paper under the same title, arguing that generative AI has inaugurated a regime of secrecy-based privatisation in which publicly sourced data become proprietary assets shielded by trade-secrecy claims (Katyal and Morten 2026). On this paper's account, the enclosure they describe is an enclosure of access, of the kinds catalogued in section 3.1, not the enclosure of governance defined here; the two analyses are complementary.

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decline, generated text may make up a larger share of the record on which future models are trained. This paper does not show that it does.

3. What is enclosed

3.1 The v1 taxonomy, retained

Version 1 distinguished four forms of enclosure of access, legal, technical, epistemic and state-enforced, and the author's later work added a fifth, institutional (hybrid) enclosure by bodies that combine the legitimacy of public service with private incentives, described as 'delegated responsibility without transparency' (Kahl 2025a; 2025b, pp 19–20). They share an assumption: that the thing enclosed exists and the question is who may reach it. That holds for weights, data and benchmarks. It does not hold for everything the third enclosure affects.

3.2 Origin, justification and authority

Debates about AI and knowledge tend to run together four questions:

- **Origin**: where did this claim come from? This has a *historical* form (what data, people and processes went into the system) and a *causal* form (what particular training material or process produced this output).
- **Evidential justification**: is the claim true or warranted, and what supports it?
- **Justification of standards**: by what standards was this claim, rather than another, selected, framed and presented, and are those standards defensible?
- **Evaluative authority**: who may make the resulting judgement consequential, and who decides whether the standards are adequate?

These can come apart. A model may generate a true proposition whose causal origin cannot be reconstructed but which a public scientific paper supports; it may reproduce, with perfect traceability, a false sentence. Origin is not justification; to treat the absence of a derivational trail as the absence of evidential grounds would be the genetic fallacy.

The distinction yields two kinds of contestation. *Propositional contestation* asks whether a claim is true or justified. *Standards contestation* asks whether the standards by which claims are selected, framed and produced are appropriate, at two levels: *output-level*, how a particular answer was produced and what it omitted; and *system-level*, whether source-selection policies, training objectives, rater guidelines, benchmarks and language coverage are defensible.

Losing causal provenance does not prevent propositional contestation in principle, but it shifts the burden of checking onto the user and leaves nothing propositional to check where the defect is an omission or a framing. Output-level contestation can draw on three kinds of reconstruction: *causal*

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(which training material contributed to an answer), largely unavailable to users beyond identifiable memorisation; *operational* (what the deployed system retrieved, ranked and called), recoverable by logging; and *behavioural* (whether answers show a demonstrable omission or bias under controlled examination), which needs neither of the others. Dissolved provenance impairs the first and leaves answers only partly explained, but it does not impair system-level contestation, since a forum can examine a provider's policies, objectives and exclusions through disclosure, testing and audit. That is why dissolved provenance aggravates unanswerability rather than constituting it.

The three operations bear on these questions without mapping onto them one to one. *Dissolution of provenance* concerns origin; *disowning of judgement* concerns the justification of standards; *concentration of recognition authority* concerns evaluative authority. The paper's central claim concerns the conjunction of unanswerability and displacement (section 3.7); the operations are enabling features that help explain why contemporary AI may produce that conjunction at scale.

3.3 Provenance: withheld, dissolved, and what dissolution costs

Withheld provenance is the familiar case: the information exists and someone holds it, such as which datasets were used and filtered, which benchmarks certified the model and what instructions guided its raters. Disclosure can repair it.

Dissolved provenance is different. A generated sentence does not come from a source as a quotation comes from a book; it is produced from parameters shaped by an enormous body of text, and nothing in the output records which texts it owes what to. Where a model reproduces memorised text, the source can often be identified. Beyond that, influence-function methods can estimate which training examples most affected an output (Koh and Liang 2017; Grosse et al. 2023), but their results are estimates sensitive to method and settings, not a recovered causal history; they are costly and designed for research; and the relevant line is between white-box estimates, made with the model's parameters and training data, and the much weaker inferences available to outsiders. Dissolved provenance therefore does not mean that causal provenance is metaphysically absent, but that it is *not individuated in the output* and can be recovered, if at all, only imperfectly and with access ordinary users lack. Nueno's image for it is the 'acousmatic' voice, speaking 'from all sources but no source in particular' (Nueno 2025); Bender and her co-authors made a related point from within computational linguistics (Bender et al. 2021).

Three evidential tasks should be kept apart: showing that material was included in training, which records and disclosure can do; showing that an output reproduces or derives from a source, which extraction and comparison can often do; and estimating how much a particular example influenced an output, which only causal reconstruction can do. Defects in training data and learned biases can often be found through audits of whole datasets, controlled interventions and behavioural testing. Causal reconstruction is therefore necessary for some questions, chiefly those about the degree of a source's influence, without being indispensable to attribution generally.

Retrieval-based systems partly restore a trail. But retrieved documents are combined with what the model brings from its parameters, answers do not ordinarily mark which claims rest on which, a

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citation may not support the sentence it is attached to, and what to retrieve is itself selected by standards the provider sets. Retrieval moves some provenance from dissolved to withheld or visible; it does not restore the reader's position with a bibliography.

The human-expert objection. Human experts also answer from memory without saying where each element came from, and professional consensus can be practically unchallengeable. The distinctive feature of generative answering is therefore not that humans answer and machines do not. Of four possible deficiencies, the absence of an *intrinsically answerable* source, of an *identified institutional answerer*, of an *enforceable obligation* to answer and of an *effective independent forum*, the first is true of every instrument and the second is partly remedied when developers publish model specifications. The argument depends on the third and fourth, combined with scale, standardisation and dissolved provenance: a small number of systems serve very large populations under centrally governed standards that answer to no forum. Dependence on others is a condition of rational belief in a complex society (Hardwig 1985); dependence on a few standardised sources whose standards no forum can test is a different thing.

3.4 In what sense these are enclosures

A natural objection is that enclosure means fencing, and that unrecorded provenance, disowned judgement and concentrated recognition authority are not fences. The objection is right about the narrow sense of the word and wrong about the concept. Ostrom's central lesson was that a commons is a regime of governance: rules, made and enforced by those who use a resource, about who may contribute and take, how disputes are settled and how the rules change (Ostrom 1990, pp 88–102). The historical enclosures extinguished customary rights and institutions of collective or locally accountable regulation and replaced them with more exclusive control. Those institutions were heterogeneous, and not everyone who depended on a commons helped to make its rules; the analogy rests on the loss of established capacities, not on an idealised form of self-government. On that understanding the three operations are conditions under which the governance of knowledge can be enclosed, and whether it is enclosed in a given case depends on displacement (section 3.7). Dissolved provenance bears a limited analogy to the extinguishing of rights of common: not to a universal entitlement to be named wherever one's work has had influence, which no one has, but to the erosion of *institutionally maintained attribution practices* where they applied, the citation that named a source, the catalogue entry that pointed to a page.

3.5 The attributive commons

What is at stake is not a body of texts, which remain where they were, but a set of practices: cataloguing, indexing, abstracting, citing, reviewing, correcting. This paper calls it the *attributive commons*. The name is analytical: it denotes a distributed infrastructure of attribution and correction whose public character lies in the circulation, persistence and reusability of contributions, not a claim that its constituent institutions were collectively governed, which must be assessed practice by practice. Attribution is its infrastructure, not a synonym for the whole: it is the rule that contributions remain traceable to their contributors, and it supports important forms of criticism and correction,

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though not all of them, since anonymous review and the checking of a proof do not depend on knowing who the author was.

Two qualifications prevent the concept from becoming nostalgic. First, it is an ideal type: an ecology of overlapping practices with different participants, rules and arrangements of accountability, controlled variously by librarians, publishers, learned societies and commercial firms. A citation convention can be collectively reproduced without being collectively governed, so the governance of each practice must be assessed separately. Second, many of its institutions concentrated authority: Dewey, subject headings, editors, peer review, encyclopaedias, the citation index and, for two decades, a dominant search engine. Every collective epistemic system has a *closure locus*, the point at which distributed inputs are stabilised into an output treated as settled, and a system can, in the author's earlier formulation, 'distribute contributions widely while concentrating closure in a small set of gatekeeping roles' (Kahl 2026a, pp 5–7): the institutions of the attributive commons often did.

What distinguished the attributive commons was plurality, and three capacities that plurality can involve must be kept apart. *Institutional contestation* is the capacity to challenge the standards of the institution that governs a practice; it is answerability if the route meets section 3.6. *Collective institutional autonomy* is the capacity of a scholarly community to found, staff and govern its own venues, such as learned societies and the journals they run, professional bodies and classification committees; it gives self-government over those venues. *Ecological independence* is the existence of viable alternatives to dominant institutions; it may constrain power without making it answerable. Autonomy is not authority over the field: a minority community may run its own journal while the dominant journals, funders and universities go on applying standards it cannot contest. Autonomy amounts to field-level governance only where the community's own institutions actually govern a material part of the practice at issue, rather than merely existing alongside institutions whose standards remain dominant. On that test some of the attributive commons was governed in common and some was not, and the strict case of enclosure applies only to the first. Both capacities, operational and constitutive, were in any case unevenly available (section 5.4).

Where the practices of the attributive commons were governed by those they served, at the level of the institution or, on the test above, of the field, they formed epistemic commons; the citation index shows that attributive infrastructure can exist without commons governance.

The attributive commons should be distinguished from the *answerable commons* proposed in the companion paper (Kahl 2026f). The first is an epistemic practice, which existed imperfectly and is being displaced; the second is an architecture of governance, proposed to secure under new conditions what the first secured under old ones.

3.6 Answerability: to whom, by what route, and a classification

Two questions must be kept apart. The first concerns a system: are the standards by which it selects and judges knowledge answerable, and to whom? The second concerns an institutional change: have such standards come to govern, or to deplete, a practice previously governed by, or answerable to, those it served? Concentration is not enclosure. A model used privately for drafting and the same

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model used to allocate public research funds have identical technical properties, and only the second need displace anything. This section answers the first question, and section 3.7 the second.

Answerable to whom. Answerability is a relation, not a property of a system. A provider may be answerable to regulators and shareholders while unanswerable to independent researchers or to a minority-language community whose literature its system passes over. Every classification is therefore made relative to a specified class of affected people and a specified standard, and membership of a class follows *exposure to consequences*. The class cannot be chosen to suit the conclusion, since for some class almost any system is unanswerable. For a finding of enclosure it is fixed by the practice said to be displaced, and the practice must be identified independently of the claim: by its function and by the governance arrangements in place before the system arrived, established on evidence, and determined by a body other than the party alleging enclosure. The practice and the arrangement said to displace it must govern substantially the same epistemic function, specified at the same level, such as one journal's editorial standards, a discipline's publishing or a system of research evaluation, not merely affect the same people; otherwise a claim could slide from governance of publication to governance of visibility or evaluation, practices the community may never have governed. The class is then those the practice served, together with its future users, represented by bodies that meet the participatory conditions below rather than by incumbent institutions as such, since incumbents may be weakened by the depletion or interested in its remedy. Findings about other classes are findings of unanswerability, not of enclosure.

The contestability condition. For a specified class and standard, the condition is met when those affected can effectively contest the standard through a route outside the control of whoever set it. Three capacities can pass for contestation: obtaining a reasoned response; securing reconsideration; and compelling a particular correction. The condition does not require the third, which no institution of the attributive commons offered. The first is not enough, since reasons can be given and criticism ignored; nor is reconsideration alone, since a standard-setter could examine objections, publish detailed reasons, refuse every change and go on applying the standard. That is *reasoned refusal*, which turns answerability into consultation. The condition is met when there is an effective institutional route through which a standard can be changed, or made to stop governing the practice, where it is found inadequately justified, by one of two routes: those affected have an *effective share in deciding* the standard; or the standard-setter is *obliged*, by an obligation someone else can enforce, to reconsider it and give responsive reasons, backed by *effective remedial authority*, the availability of a binding consequence such as disapplication, a binding obligation to revise, enforceable suspension of reliance, or withdrawal of a decision reached under the standard. Answerability does not guarantee that any particular challenge succeeds, and who decides whether a standard is inadequately justified differs by route: under the first, the community, by its own rules; under the second, the forum. A participant who exercises an effective share in deciding and loses has not met unanswerability, and disagreement with an outcome is not evidence of it. A standard-setter that changes its practice under public pressure or competition, but at its own discretion, does not meet the condition: responsiveness is not answerability. The condition institutionalises and strengthens Longino's requirement of uptake (section 3.8); the binding element is the paper's own. The concept is *effective epistemic answerability*, deliberately more demanding than ordinary accountability, which

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explanation, reconsideration and the correction of procedural error can satisfy. The reason for the higher threshold is its object. The question is not whether an institution must explain its conduct, but whether those affected keep an effective means of preventing an inadequately justified epistemic standard from continuing to govern them, and a forum that can identify defects but cannot affect the standard does not preserve that means. Nor need it substitute a standard of its own: authority to disapply, to require lawful reconsideration or to suspend reliance is enough. The threshold does not condemn self-governing academic judgement. A university that reserves academic judgement to its own bodies meets the first route where those bodies give the people affected an effective share in setting the standard. What the test excludes is the case in which the only route is review confined to legality, rationality and procedure; such an arrangement is accountable for decisions but not answerable for the standard, and the classification is deliberate. The route must also be *external*: a developer's internal red-teaming or ethics review may be rigorous, but it is criticism the developer commissions and may disregard. It is scrutiny, not answerability.

The concept differs from the contestability discussed in work on algorithmic systems, which has centred on challenges to individual decisions and on participation during design. An analysis of submissions to an Australian consultation found contestation understood chiefly as the challenge of decisions after the event, alongside calls to extend it to system design and to collective routes (Lyons, Velloso and Miller 2021); frameworks of 'contestability by design' extend it across the life cycle of a system, including stakeholder participation, scrutiny of the system as a whole, collective action and third-party oversight (Alfrink et al. 2023). The concept here shares their concern with system-level scrutiny but takes the governing standard as its object, assessed relative to a specified class and, where governance lay there, at the level of a field, and it requires an external route with binding effect on the standard. On that test, participatory practices conducted by the developer are scrutiny unless they carry a binding consequence the developer does not control.

Two routes, two goods. One route is an *independent forum*, a court, regulator, ombudsman or arm's-length body that can require reasons, oblige reconsideration and impose a binding consequence on a standard it finds inadequately justified. Its jurisdiction must reach the adequacy of the standard for the function it governs, not only procedural compliance, consistency of application or bare irrationality, since a forum confined to those grounds leaves the standard itself insulated. The test asks whether such a forum exists and can effectively be used, not whether the standard is in fact justified: enclosure turns on whether a standard's justification can be effectively contested, not on its merits, and where reasonable standards compete, a forum that upholds the provider's choice has still made it answerable. The other is a *collective-choice arrangement* in Ostrom's sense, in which most of those affected by the rules can take part in changing them (Ostrom 1990, p 90). Longino's first condition of objectivity is recognised *venues* for criticism, in the plural (Longino 2002, ch 6); it does not require a tribunal. *Critical answerability*, the assurance that objections receive reason-responsive consideration with effect, gives institutional and binding form to the uptake of criticism that Longino's account requires; *participatory authority*, an effective share in setting the standard, is what makes an arrangement a commons. Neither guarantees the other, and they protect different goods: participatory authority protects collective self-government, critical answerability justified contestation. Both count as answerability because each gives those affected effective control over

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whether an inadequately justified standard goes on governing them, but they do so differently. Under the first route the community judges the merits itself, so a self-governing community is not dispossessed when some participants lose legitimate collective decisions; enclosure protects effective collective control, not an individual veto. Systematic exclusion defeats the first route for the excluded class, because that class does not genuinely share in the authority said to govern it; the conditions below are what show that it does. Both need *institutional standing*, so that objections enter the process as of right, and *effective uptake*, so that they can change the outcome. Participatory authority further requires meaningful representation of those affected, access to the information on which standards are set, rights to make proposals and set the agenda, and protection against systematic exclusion. A practice in which those it serves hold participatory authority, at the level of the institution or of the field, is an *epistemic commons*; a practice answerable to them through either route is *contestably governed*.

The trail condition. *Source-level causal attribution* is available: the sources that materially contributed to a particular output can be identified. Attribution here means a defensible causal attribution under a stated method, and the classification is relative to a specified attribution task: the influence of training sources, the sources used at inference, or both. Training examples, retrieved documents, system instructions and tool results affect outputs through different mechanisms, and a system may meet the condition for one task and fail it for another. A logged retrieved document or a displayed citation shows what was available or offered as support, not what materially caused the output, and does not by itself satisfy the condition. This is narrower than a complete causal explanation, which even full access to weights, data and logs would not necessarily provide, and it is an ideal type, applied to a specified class of outputs. When the contestability condition fails for an affected class, the standard is unanswerable to that class; when the trail condition also fails, the unanswerability is compounded. The four combinations classify relations between a system, a standard and an affected class, not systems:

	Standards effectively contestable	**Standards not effectively contestable**
Source-level causal attribution available	A. *Contestable concentration*	B. *Unanswerable concentration*
Source-level causal attribution unavailable	C. *Causal provenance unavailable*	D. *Compound unanswerability*

The same system can be in A for one class and in B for another. The table measures *institutional answerability*. A second property belongs to the field: *ecological contestability*, whether those affected have effective alternatives through which to challenge, bypass or counterbalance a standard. It may rest on two different things, not stronger and weaker forms of one. *Collective institutional autonomy* is a form of governance: where the community's own institutions govern a material part of the practice (section 3.5), it is participatory authority exercised at field level and counts for the purposes of section 3.7. *Ecological independence* is a condition of the institutional environment, the availability of alternatives; it does not count, though its loss is still a harm of concentration.

Configuration A is concentrated but contestable. Peer review is in A relative to those whose participation rights meet the conditions above, such as members of a learned society that governs its journals and chooses their editors. Ordinary editorial discretion earns no presumption of

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contestability, and relative to early-career researchers, independent scholars and those outside dominant institutions and languages, who are subject to its standards without effective influence over them, peer review is in B. The third enclosure intensifies an older problem; it does not introduce unanswerability into a world that was contestable for everyone.

Configuration B is unanswerable concentration with the trail intact. Ranked search preserved the trail, but its ranking standards were secret and changed at the firm's discretion. The Science Citation Index preserved the trail completely, and its standard of source selection responded to the community in one way only: ISI selected source journals by the citations they received, so that, as Derek de Solla Price wrote in 1979, the list of cited journals provided 'a higher criticism of which journals to accept and which to reject as sources' (quoted in Wouters 1999, p 113), and the adequacy of the source journals was confirmed by measuring how often they were cited (Wouters 1999, p 114). For publishers and fields that thought the standard wrong, the one route tried, an antitrust complaint, failed, and no other is reported that could oblige ISI to reconsider it (Nueno 2023, p 138).

Configuration C is the unavailability of causal provenance without unanswerability, whether provenance is withheld or dissolved (section 3.3). The label names a lack of source-level causal attribution, not a lack of documentary traceability: a system in C may give careful citations and full retrieval logs. Propositional, operational and behavioural contestation would remain available, and the loss would transfer no governance. It matters chiefly for what it does to detection. Where the contestability condition fails, dissolved provenance makes the failure hard to see from outside: no output points back to the standards and sources that shaped it, so exclusions show up only through deliberate testing, and those excluded may not learn that they were. That is the work provenance does in the theory. It is not a condition of enclosure, since transfer can occur with full provenance, but it is a reason why configuration D persists.

Configuration D is compound unanswerability. It describes many dominant general-purpose answering services as currently deployed, relative to independent researchers, ordinary users and the communities their answers leave out, in respect of their system-level standards; deployments under externally imposed standards or institutional oversight may be in A or C for the classes those arrangements protect. D is not new; what is new is that it is a configuration of general-purpose public knowledge infrastructure. Whether a system in D encloses anything depends on what it displaces.

3.7 Displacement and enclosure

A provider's standards, however unanswerable, govern only its own system. A practice is enclosed only when those standards come to govern it in place of those it served, or when reliance on them depletes it.

Reliance and delegation. In *epistemic reliance*, an individual treats an output as credible, and transfers no authority over anything but her own belief. In *institutional delegation*, an organisation makes an output presumptively authoritative within an established decision process, as when a funder treats a model's assessment as the default. Neither is yet enclosure. A library could hand its reference desk to a commercial system while keeping authority over its sources, audits and withdrawal, so that

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the function moves but not its governance; a university could keep its own staff evaluating research while requiring them to follow a proprietary ranking, so that the work remains human but evaluative authority moves.

Two mechanisms of displacement. Displacement occurs by two mechanisms, species of one genus: the material displacement of a community's effective capacity to sustain and contest the epistemic practices on which it depends. The capacity to sustain has two parts. *Operational capacity* is the ability to go on producing, correcting and preserving public contributions; *constitutive capacity* is the ability to found, sustain and govern the institutions and standards through which contributions are recognised. The account rests on an explicit premise: *a community's effective control over an epistemic practice includes the material conditions needed to sustain it*. A formal right to govern a forum that can no longer attract contributors is not effective control, as a formal right of contestation that cannot be used is not effective contestation; the paper judges capacities by their effectiveness throughout. A community therefore does not retain effective governance of a practice merely because it retains formal authority over its rules, if either capacity has been materially displaced into an arrangement it cannot govern. Transfer threatens control over standards; suppression can threaten either capacity. The historical analogy is stronger for transfer than for suppression. Transfer, like the enclosure of land, dispossesses a community of authority over a practice; suppression extends the concept from dispossession of governing authority to displacement of the material capacity to reproduce a governed practice. That extension is the paper's own theoretical step, not an implication of the history. To the objection that a community which drifts away from its forum has lost an activity, not been dispossessed of a right, the answer is that voluntary migration alone is depletion; what makes it a loss of control is the conjunction of dependence on a substitute the community cannot contest with the disappearance of the contestable public function the practice performed. Depleting operational capacity impairs constitutive capacity when the contributors, expertise and resources it drains are those a community needs to sustain or reproduce the venues through which it actually governs a practice.

- *Transfer* removes the capacity to contest. It requires two things together: the standards that effectively govern an epistemic function pass from a contestably governed practice to a system whose standards are unanswerable to those the practice served; and those affected have no effective route to stop that standard governing the practice, whether by challenging the provider or by compelling the institution that adopted it to abandon it. An external standard that cannot be challenged at its source does not enclose a practice that keeps effective control over whether to use it. Enclosure is the loss of control over *whether* a standard governs, not the use of a standard no one can challenge. A right to have individual decisions overturned is not by itself such a route: those assessed may win particular appeals while the standard goes on governing every case not appealed. What counts is whether reliance on the standard itself can be stopped, the distinction between answerability for conformance and for the standard (section 4.3).

- *Suppression* removes the capacity to sustain. When people put their questions privately to a model rather than publicly to a forum, the model does not take over the forum's governance; what changes is the activity. The forum governed a public contribution, posted, voted on,

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corrected and kept for others; the model governs a private interaction that produces nothing public. Suppression comes about through individual reliance at scale, without anyone deciding anything.

Because depletion is common and often benign, suppression counts as enclosure only where three conditions hold together:

1. the displaced practice performed a demonstrably valuable *public epistemic function*, such as the public questioning, correction, persistence and reuse of knowledge, and was *contestably governed* by the community it served, at the level of the institution or of the field (section 3.6), though not necessarily as a commons;

2. the replacement does not preserve a *functionally adequate, publicly contestable* means of performing that function, in the substitute or elsewhere; and

3. the loss is *attributable*, to a material degree, to substitution by an intermediary whose standards are unanswerable to those the practice served: the *immediate epistemic service* the practice provided, such as obtaining an answer, migrates to that arrangement, or to a class of such arrangements, on which the community, including its future users, comes materially to depend, while the *public function* the practice generated is impaired.

Without the first, the decline of a forum of no public value would count, and so would the decline of a service never answerable to those it served, which displaces no governance; without the second, a practice replaced by a better public one; without the third, depletion through poor moderation or migration to an answerable system, or the decline of a forum whose users only occasionally consult a service they cannot challenge. *Material dependence* is a practical constraint on exit, not high usage: a community depends on an arrangement, or a class of them, when it cannot obtain the immediate service through a functionally adequate alternative outside that class without disproportionate cost, loss or disruption. Widespread use with easy exit is reliance, not dependence. That an intermediary's standards come to determine what counts as an adequate answer is a different matter, which bears on transfer. Future users matter here because they may inherit a depleted public infrastructure and face fewer alternatives than those who first chose the private service. Dependence is required because it is what turns depletion into a second-order loss: the community has lost an independently governed public function and come to rely, for the service that replaced it, on an arrangement whose standards it cannot control. Where an answerable substitute is available, as with a public answering service whose standards are independently contestable but whose exchanges leave no public record, that second loss is absent although the public function remains depleted; the case is depletion, to which the externality ground of section 6.3 still applies. That classification is a substantive choice, which the definition makes through its requirement that displacement occur through a system whose standards are unanswerable. A broader theory could count sufficiently irreversible destruction of the material conditions of community governance as dispossession whatever the substitute; this paper declines to, so that not every serious externality becomes enclosure. Dependence may attach to a class of intermediaries rather than one provider; otherwise competition among unanswerable providers would defeat a finding of enclosure even where together they had displaced a public

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practice. The kind of public loss can be the same whether the substitute is answerable or not, though its size and persistence may differ, since an answerable substitute can be made to restore what an unanswerable one cannot; the conjunction adds a second loss: the public function does not migrate but disappears, and the service that replaced the practice, on which the community comes to depend, lies beyond its control. Depletion is an externality; depletion into an unanswerable substitute on which the community depends is enclosure. Making the substitute answerable ends the condition of enclosure, not the loss it caused: the depletion remains, and repairing it may require the reconstruction of institutions (section 3.8). What suppression protects is the continued availability of a valuable public function under contestable governance, not the survival of the practice that once performed it: a better private service that leaves the public function available elsewhere is not suppression, and the theory is no presumption against beneficial substitution. At the level of a field, constitutive capacity is protected insofar as it is established and materially exercised: the institutions through which a community actually governs a material part of the practice (section 3.5), and the contributors, expertise and resources needed to reproduce them. Its material depletion can be suppression under these conditions. A reduced prospect of founding further institutions, while existing ones still perform the function adequately, is a loss of resilience and may be an early sign of enclosure, but it is not suppression; otherwise the concept would become a general theory of lost institutional opportunity.

Both mechanisms must be material and are matters of degree, so that enclosure can be partial. They do not carry the same evidential burden. Transfer can be shown from decision procedures, appeal rights and decisions. Suppression requires a counterfactual, and the evidence must distinguish loss of activity from loss of value, temporary migration from persistent depletion, decline of an institution from loss of its function, dependence on a service from dependence on its standards, and change induced by AI from change that would have occurred anyway. Neither is specific to AI: a private tutoring service, answerable to no one, that came to answer most of a field's questions in private could suppress a public forum in the same way. AI matters because a few systems combine very wide reach, standardised answers, synthetic presentation, dissolved provenance and private interaction, so that substitution a human intermediary could achieve only locally could become material across a public.

When an external standard governs. An institution can rely heavily on an external measure while keeping its own standards in charge. A standard is *governing* rather than merely *evidential* when it is presumptively authoritative, when departures from it require special justification, when rejecting it carries practical penalty, or when it determines outcomes independently of other evidence, whether by written rule, by expectation and penalty, or in effect; effective control over outcomes is the strongest evidence of transfer.

Gateways. Suppose a dominant answering system becomes the principal route to a field's work while passing over many of its journals, say those of a minority-language tradition. The case is *transfer* where the gateway's selection becomes governing, as where institutions treat the set it surfaces as the relevant literature in consequential decisions and those affected cannot have that reliance stopped. It is *suppression* where the passed-over journals, or the field's organised capacity to sustain them, lose

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readers, contributors and resources until their public function is materially impaired under the three conditions. Otherwise it is structural recognition displacement (section 5.3), a harm of concentration but not enclosure. Counting that too would make the reduction of alternatives, which every successful product achieves to some degree, sufficient for enclosure.

Enclosure. The definition can now be stated. The protected object is *a community's effective capacity to sustain and contest the epistemic practices on which it depends*. *Enclosure* is the material displacement of that capacity, by transfer or by suppression, through a system whose standards are unanswerable to that community. *Strict enclosure* displaces an *epistemic commons*, a practice in which those it serves hold participatory authority, at the level of the institution or the field; it keeps the historical structure, in which participants are dispossessed of governance rights. *Enclosure in the wider sense* displaces a *contestably governed practice* whose standards those it serves can contest through enforceable rights without a share in deciding them, as with a public archive whose users can challenge its selection standards before an independent body. The wider sense is an extension justified on normative and functional grounds, not by historical identity: both forms are kinds of effective community control over the conditions of epistemic practice, self-government in the one case and accountability in the other. Both senses are relative to a class. A practice may be an epistemic commons for its members and unanswerable to non-members who depend on its functions, so a claim of strict enclosure can be made only on behalf of a class that held participatory authority, and a claim in the wider sense only on behalf of a class that held effective contestation; otherwise a claim of enclosure would reproduce the exclusions of the institutions whose displacement it laments. The history shows only that commons governance need not be direct, since rights of common were often enforced in manorial courts rather than made by the commoners, and Ostrom's design principles include conflict-resolution mechanisms and monitors accountable to users (Ostrom 1990, p 90). Membership of the wider category turns on the effectiveness of a right of contestation, not its source: a general public-law right of review counts where the community a practice serves can effectively use it.

In one line: *unanswerable standards + material displacement, by transfer or suppression = enclosure*. The three operations are contributing conditions, not alternative definitions.

How the definition can fail. A finding of enclosure fails if the evidence is absent. For transfer, the decisive evidence is a change in who has effective authority over the standard governing an established function, together with evidence that those affected cannot have its use stopped. For suppression, it is the loss of a valuable public epistemic function that was contestably governed by those it served, compared with what would otherwise have existed and attributable to substitution, together with material dependence on an unanswerable substitute for the immediate service. Where either element is missing, the case is concentration, influence or depletion.

Applications. The test gives graded verdicts, argued where the evidence is discussed. The citation index's metrics were incorporated into research evaluation and amounted to transfer only where they became governing and those assessed could not stop their use (section 5.1). Ranked search was unanswerable but, on the evidence offered here, displaced no practice governed in common. Public

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question-and-answer forums show a plausible mechanism of suppression, not suppression in the defined sense (section 6.3); where public exchange is depleted but the replacement remains optional, or the function was never contestably governed, an obligation to sustain public capacity can arise without enclosure, which is a result of the test, not a gap in it.

The contemporary sequence. In the contemporary case the mechanism typically runs through five steps. The second is characteristic of the present configuration rather than necessary, since publicly held systems can enclose as well:

1. *Distributed epistemic contribution.* Authors, editors, reviewers, librarians and the users of public forums produce and organise knowledge.

2. *Proprietary consolidation.* That material is gathered into systems that are privately held and governed.

3. *Constitution of system-level standards.* The provider settles the standards by which the system selects and answers.

4. *Delegation or reliance at scale.* Institutions make the system's outputs presumptively authoritative, or users rely on them in numbers large enough to withdraw contribution from public forums.

5. *Displacement.* A community's capacity to sustain or contest a practice on which it depends is materially displaced, by transfer or suppression, under standards in configuration B or D.

Only the fifth step is enclosure; the first four are its conditions. Proprietary development is therefore not by itself an enclosure of governance.

3.8 Why contestable closure matters

A critic could accept every descriptive claim and reply that competent unilateral governance might produce better answers than a contestable commons. The paper needs a reason to value contestability.

The best-developed reason is Helen Longino's. Objectivity, on her account, is a property of communities rather than individual knowers, and a community's inquiry is objective to the degree that it has recognised venues for criticism, uptake of criticism, public standards by which criticism is assessed, and tempered equality of intellectual authority (Longino 1990; 2002, ch 6). Longino's conditions apply to epistemic *practices*, not to individual outputs or systems: a model in a community whose institutions of criticism remain strong does not by itself make its knowledge less objective. The argument is therefore conditional. *Where* systems in configuration D become a community's dominant intermediaries, and other institutions cease independently to scrutinise their outputs and standards, the social conditions of objectivity are weakened. The three operations do not undermine Longino's conditions by themselves: an answer can be tested without its causal provenance, standards can be documented even where responsibility for them is disowned, and concentrated recognition need not defeat tempered equality within communities that inquire independently. They undermine

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those conditions when they contribute to displacing the independently governed venues in which criticism, public standards and tempered equality are sustained. The claim concerns conditions, not outcomes, since a community that loses some venues of criticism may gain others.

Longino gives a reason to value contestable practices; she does not by herself supply enforceable duties. The step needs a premise of its own, which the companion paper develops (Kahl 2026f):

> *Where an intermediary becomes materially consequential to the conditions under which a community produces, tests and recognises knowledge, its governing standards acquire a public significance that warrants obligations of answerability proportionate to that consequence.*

The premise is technology-neutral, and consequence gives only a *prima facie* reason for answerability, not a licence for any particular form of it. It follows that enclosure is a particular failure of epistemic governance, not the general condition that justifies duties of answerability. A consequential, unanswerable system governing a function that never existed before would warrant such duties without enclosing anything, since nothing previously governed would have been displaced.

The concept nonetheless does work that its two components do not do separately. Its distinctive object is a *second-order loss*. An unanswerable standard can produce bad decisions; when its adoption displaces the institutions or capacities through which it could have been resisted, the community loses its capacity to make, contest or replace the standards that govern it. The second-order loss is not a third condition, to be proved separately: it is what unanswerable standards and material displacement together amount to, and the reason the conjunction matters. The harm then tends to reinforce itself: on the hypothesis the citation-index case illustrates, the institutions that could have exposed a displacing standard lose participation or authority, and evidence of harm becomes harder to obtain the longer it governs. Two consequences follow, which the companion paper develops. Where enclosure is in prospect, investigation may be warranted before harm is fully shown, in proportion to the gravity and likelihood of the harm and the costs of investigating, though binding measures are not; and where enclosure has occurred, correcting a defective standard is not enough, since what was lost is a capacity. Binding contestability is, moreover, an institutional safeguard against unanswerability, not a guarantee of objectivity. An independent forum can itself become a closure locus, drawing on limited expertise or on a dominant school's assumptions, and that is a limit of the second route to answerability that any proposal for such a forum must meet.

Three grounds for duties can then be kept apart, overlapping but not entailing one another: consequential epistemic intermediation, which grounds answerability for standards; provenance of contribution (section 6.2), which grounds preservation of attribution and standing for contributors; and a substitution externality (section 6.3), which grounds a duty to help sustain public functions.

Robert Merton's norm of 'communism' in science states the attributive commons in its classic form: findings are assigned to the community, and the scientist's claim on them is limited to recognition and esteem (Merton 1973).[3] The third enclosure inverts both terms. Findings are consolidated into

[3] Merton's essay was first published in 1942 as 'Science and Technology in a Democratic Order'.

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privately governed systems, and recognition, the one thing the contributor was owed, depends on attribution practices that such systems need not maintain.

4. The fiction of unmediated judgement

4.1 Judgement relocated, not removed

The promise of the statistical order of knowledge was that the indexer could be removed. Nueno's history shows that the indexer was moved: producing each edition of the index involved 'constant decisions' at every stage, from the selection of journals to the consolidation of references, which ISI's claims to objectivity concealed (Nueno 2023, pp 137–138), and 'subjective or human aspects have only been displaced into the design of the parameters' (Nueno 2023, p 34).

Wouters's history shows that removing judgement was a condition of the index, not a side effect. Garfield first tried to give citations a classification like the one Shepard's Citations used for legal cases, marking whether a later decision affirmed or reversed an earlier one. Joshua Lederberg told him that this would be self-defeating, since abstracting services that judged the literature already ran years behind because they had to make those judgements. Automation, Wouters concludes, 'was, however, only made possible by stripping the citation of every judgment on its meaning' (Wouters 1999, p 74), and Lederberg presented this as a virtue: 'The brilliant utility of the citation index approach is that it cuts across the problem of meaning by an automated procedure' (letter to Garfield, 9 November 1962, quoted in Wouters 1999, p 74). What remained was a count, whose meaning was supplied later by whoever used it to evaluate, in a form that could present itself as no judgement at all. In 1964 the documentalist Mortimer Taube objected that no amount of quantitative manipulation would let anyone avoid 'the responsibility of evaluating individual contributions' (Taube 1964, quoted in Nueno 2023, p 135). Quantification does not remove judgement. It changes where judgement is exercised and makes it possible to deny that it is exercised at all.

4.2 Neutrality claimed, judgement retained

The denial does practical work. Nueno observes that platforms have argued against responsibility for what they display on the ground that they perform no editorial or indexing function, and that 'as long as the fiction that algorithms organize information automatically and without human mediation persists', they can present themselves as mere containers (Nueno 2023, pp 210–211). In other legal contexts, and in respect of different activities, the same kinds of actor claim editorial freedom when their selections are regulated. In the United States, for instance, a provider is not treated as the publisher of information provided by another (47 USC § 230(c)(1)), while the Supreme Court, vacating and remanding decisions on facial challenges, indicated that curating what to include, exclude and prioritise is protected expression (*Moody v NetChoice*, 2024). The two rules answer different questions, and whether either reaches what a provider's own model generates is unsettled; the point here is only that an actor can claim the benefits of automated neutrality while retaining

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control over the criteria of selection. The denial that any judgement is exercised, where it reduces particular costs of accountability, is what this paper calls the *disowning of judgement*. The legal position in four jurisdictions is examined in the companion paper (Kahl 2026f).

4.3 Answerability for the standard

When an institution controls not only a decision but the criteria, evidence, forum and remedies through which it could be challenged, accountability mechanisms exist on paper and are empty in practice (Kahl 2026b); the disowning of judgement places the first two inside a system said to decide nothing. A second distinction concerns the object of answerability. An agent who applies a standard can be answerable *for conformance*: did this output meet the standard? An agent who settled the standard can be answerable *for the standard*: why this standard and not another? These may end at different agents, because a standard is settled at one point and applied at many (Kahl 2026d). Some inherited structures are *epistemically constitutive*: a training objective, a reward model, a labelling guideline or a benchmark specification does not supply a rule its user could read and resist, but the space within which anything can be recognised as relevant. And some dimensions of a standard accrete without anyone settling them: someone decided what a benchmark measures, but no one decided that it should be the measure.

A ranking system or language model embodies many such standards: what counts as relevant, harmful, helpful, authoritative or adequate. Each was constituted somewhere, yet the public account typically locates behaviour in the mechanism, the model learned it, the algorithm surfaced it, and so makes everyone in the chain answerable for conformance at most. The standards are also not all epistemic. A source may be selected because it is reliable, but also because it is licensable, cheap to process, safe to display or commercially advantageous. Each consideration can be legitimate; the problem arises when mixed criteria govern outputs treated as knowledge, so that non-epistemic considerations acquire epistemic authority without being disclosed or justified to those affected.

4.4 Judgement in the language model

A ranked index returned a list, and the judgement behind it was partly visible. A language model typically returns one text, in which selection among sources, the weighing of conflicting claims and the resolution of ambiguity have happened before the user sees anything; some systems expose competing sources or their uncertainty, and the point concerns the typical presentation of a synthesised answer. Behind the text stand identifiable decisions about training data, filtering, objectives, rater guidelines and evaluation, much of it carried out by workers whose contribution is, in Mary Gray and Siddharth Suri's phrase, 'ghost work' (Gray and Suri 2019).

The model itself is not the answerer for this judgement. Authorship of a judgement turns not on how fluently a framing is used but on how it came to hold its authority: whether it was ever brought before the agent's own evaluative capacities and retained through their exercise. For standardly built systems, the governing frame's authority 'was fixed by training and objective-setting', with no process amounting to the system's appropriating it (Kahl 2026e, p 34): the system was never in a

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position to assess the frame, so it cannot answer for having accepted it. The standards a model applies were therefore constituted upstream, by people and processes that can be identified, or accreted in a way someone must be made responsible for. The gap is in the public account and the institutions around it: describing the model's behaviour as something it 'learned' or 'decided' removes the constitutors from view, and the absence of a forum keeps them there.

5. Recognition engines

5.1 From search to evaluation, and the citation index as a case

Nueno's chapter on the Science Citation Index is titled 'Engines of Inequality'. Built to help scientists find literature, the index was adopted as an instrument for measuring impact when post-Sputnik funding created a demand for numbers that could justify allocation; by pairing results with counts it rewarded those at the top and pushed others out of view, and it helped establish the dominance of Anglo-American research and of English by under-representing work in other languages (Nueno 2023, pp 33–34, 121–123, 206).

Wouters's history supplies a sharper distinction. Following Derek de Solla Price, Wouters distinguishes the *reference*, which belongs to the citing text, from the *citation*, which the index produces by turning lists of references 'upside down', so that each reference becomes an attribute of the cited text; the inversion is 'the basic symbolic act of producing a citation index' (Wouters 1999, p 11), and the citation 'is the product of the citation indexer, not of the scientist' (Wouters 1999, p 4). Because only citations an index makes available can be counted, 'being cited is not exactly the same as receiving a citation' (Wouters 1999, p 12). The index did not merely collect acts of attribution; it reassigned them, and made the recognition a work could obtain depend on whether an indexer had inverted the references to it. The currency of the attributive commons became the currency of a ranking. The shift in evaluative authority passed through four standards set by different parties: what entered the index and how references were counted, both set by ISI; what the counts were taken to signify, shaped jointly by ISI, bibliometricians and adopting institutions; and how institutions used the measures. The mechanism this paper relies on is narrower than Nueno's title: the index converted existing advantages in visibility and citation into institutional advantage in evaluation.

The Dutch case Wouters reconstructs shows a provider's standard and institutional delegation working together. In 1983 the Netherlands' Advisory Council for Science Policy (RAWB) issued a report on priorities in health research strongly based on data from the Science Citation Index, combined with expert opinion, so that expert opinion was no longer the sole source of quality assessment (Wouters 1999, p 153). Because the council advised government, its use of citation analysis was politically accountable: sharply criticised, it had to respond, and the government partly distanced itself from the report (Wouters 1999, pp 158–159). But when critics objected that not all journals had been included, the council's secretary replied that 'for reasons of quality assurance we have only used SCI journals' (Wouters 1999, p 155). The reasons ran out at ISI's selection standard,

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which the council could invoke but not answer for. Transfer is not established. The council kept expert opinion beside the counts and was free to discount them, which is evidence against complete delegation, though not conclusive evidence against practical presumptive authority, since a measure can govern in practice while expert opinion formally remains available. Three stages should be kept apart: *incorporation* of an external standard, its acquisition of *presumptive authority*, and *transfer* of effective control over the governing standard. The evidence establishes incorporation, suggests the possibility of presumptive authority and does not establish transfer. Whether the council's use was contestable in the demanding sense of section 3.6, as distinct from politically accountable, the evidence does not show. The citation index is therefore a precursor that reveals the mechanism by which a privately constituted standard can acquire institutional authority, not a proven instance of the third enclosure.

Evidence that ISI's metrics later acquired institutional endorsement is stronger; evidence that they became governing remains incomplete. A study of universities' review, promotion and tenure documents found that 40 per cent of research-intensive institutions mentioned the journal impact factor (an ISI product) or closely related terms, that 87 per cent of those that mentioned it supported its use, and that none prohibited it (McKiernan et al. 2019). That is evidence of formal endorsement, which goes beyond influence towards presumptive authority, but mention is not use, support for a metric's use does not show that it controlled outcomes, and where a university made the impact factor governing, the practice remained contestable if those assessed could have the university's reliance on it stopped. The reform movement corroborates the concern: the San Francisco Declaration on Research Assessment recommends against using journal-based metrics 'as a surrogate measure of the quality of individual research articles' in hiring, promotion and funding (DORA 2012), and the Agreement on Reforming Research Assessment commits signatories to abandon inappropriate uses of journal- and publication-based metrics (CoARA 2022), though such declarations show that misuse was perceived to be widespread, not how widespread it was. The metrics amounted to transfer, and to enclosure, only where they became governing and those assessed had no effective route to stop their use. Where peer judgement was governed by those it served, through a learned society or the organised capacity of a field to set its own standards, such cases approach strict enclosure.

The mechanisms that followed are well documented (Merton 1968; Strathern 1997; Espeland and Sauder 2007; Porter 1995).

5.2 The recognition game

Metric governance reshapes knowledge production by intensifying competition for recognition. Competition produces rivalry and what the author has called *epistemic clientelism*: researchers moderate their judgement to remain acceptable to those who control advancement. Metric evaluation *tends to privilege* contributions already legible to established patterns of uptake, especially over short assessment horizons, because a count rewards work in proportion to its uptake by those already working within established programmes, though how far it does so depends on the metric and its use, and citation measures can reward disruptive work, if slowly and unevenly; and when a metric is treated as a verdict, responsibility for misrecognition becomes diffuse, *answerability laundering*,

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since each party can point to the number and none to a judgement (Kahl 2026c). The Science Citation Index created conditions in which these mechanisms could operate. Recognition was already scarce; the index made certain dimensions of it commensurable, rankable and administratively allocable, and so may have intensified competition for scarce institutional rewards. Nueno associates it with 'a new scientific self', a scientist-entrepreneur who could publicly appear 'motivated by competition, contest, and reward' (Nueno 2023, p 114). Whether epistemic clientelism and answerability laundering occurred in the ISI case specifically is not established by the literature cited here; the claim is limited to the conditions the index created, in which a count could decide where no one had to say why. Its commercial successors sell ranking and evaluation data to governments and agencies (Nueno 2023, pp 207–208; Mirowski 2011), and Mario Biagioli and Alexandra Lippman argue that the impact factor 'is about creating the conditions of possibility for a market' (Biagioli and Lippman 2020, p 7; see Nueno 2023, p 208). The commons ISI drew on was the practice of citation. What it sold back was recognition.

5.3 The common mechanism: delegation and reliance

Language models bear on what counts as a *source*, an *answer* and an *author*. Source recognition concerns epistemic visibility; answer recognition concerns standards of adequacy; authorship recognition concerns the attribution of responsibility and originality. That a system influences all three does not show that it governs them.

The common mechanism, where there is one, is the formation of evaluative authority outside the community whose recognition is at stake. In the institutional form, an institution delegates evaluative discretion, making a system's outputs presumptively authoritative within its decision process. In the diffuse form, no one assigns authority to the system, but enough individuals rely on its outputs that they acquire authority in practice. Delegation is a matter of degree, and this paper speaks of *institutional recognition capture* only where three conditions hold together: *presumptive authority*, the system's assessment being the default unless rebutted; *institutional consequence*, the assessment affecting someone's opportunities, standing, publication or recognition; and *deficient recourse*, the person affected having no effective route to challenge the governing standard or its application (section 3.6). Such capture is one route through which transfer may occur, but is not by itself sufficient for enclosure. The sequence runs from recognition influence through institutional capture to possible transfer, and reaches enclosure only where prior effective governance of the practice is displaced. If a dominant answering system systematically passes over a minority-language literature and becomes the main route by which the public and other researchers find work, it alters who is recognised without any decision about any scholar: *structural recognition displacement*, recognition redistributed through reliance at scale. It may contribute to suppression under the tests of section 3.7, but need not, since researchers may go on judging the omitted work by their own standards. *Recognition influence*, a system's effect on what is noticed, credited or accepted, is the weakest form. The *concentration of recognition authority* names the tendency common to all three.

Sources and answers. A model's citations are shaped by what it can retrieve and by the answer it is composing, so that works not digitised, not in dominant languages or behind barriers it does not cross

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do not appear at all; and a generated answer performs the judgement of adequacy a ranked list left to the user. Both are influence, and become institutional capture when an institution treats the surfaced set as the relevant literature, or the system's answers as its measure of adequacy, in consequential decisions about people.

Authors. In autumn 2026 the Académie Goncourt withdrew Thélyson Orélien's novel *C'était ça ou mourir* from its prize selection, citing plagiarism and the 'convergent results' of analyses suggesting it was 'in all likelihood very largely the product of an artificial intelligence' (Ageorges 2026, translated). Detector outputs varied widely (AFP 2026), and his publisher later ended its relationship with him on the strength of a metadata analysis, while he maintained that he had used AI only for research (Glorieux-Stryckman 2026). Human institutions decided, and neither relied on a classifier alone: automated analyses acquired evidential weight, not authority. Even accepted, such evidence answers a question of production, not authorship, since whether a text was produced by an agent's cognition differs from whether the agent is answerable for it as his or her own (Kahl 2026e, pp 3–5).

5.4 Inherited standards

Recognition influence may reach people who never use the system. Three propositions must be kept apart: AI-assisted writing changes the characteristics of published text; those changes shape the expectations of editors, reviewers and readers; and writers who do not use AI adapt their work to meet them. The first is supported: an analysis of more than 15 million biomedical abstracts found an abrupt post-2022 rise in particular style words and estimated that at least 13.5 per cent of 2024 abstracts were processed with a language model, up to 40 per cent in some subcorpora (Kobak et al. 2025). The second and third are hypotheses, plausible but not, so far as this paper is aware, established.

If they hold, writers who never use AI would reason under standards of adequacy that AI systems helped to set. The author's account of authorship distinguishes framings an agent has *appropriated*, by bringing them under genuine assessment, from framings *available but unappropriated*, which are 'genuinely hers to reason under without being hers to answer for' (Kahl 2026e, pp 30–31): she has not examined them, so cannot be held to account for them, yet they shape what she can say. Standards absorbed from the ambient texture of a literature would usually be in that second state. That alone is the ordinary human condition, and the plurality of pre-AI conventions was never available to everyone: academic English and metropolitan publishing standards were already concentrated. What would distinguish AI-shaped standards is the position of whatever stabilises them. A dominant system largely reproduces distributed practice, but it selects among competing conventions and amplifies some at the expense of others, through objectives, rater guidelines and evaluation procedures that the inheritor cannot see and no external route can correct. The claim is therefore that AI may accelerate and standardise existing exclusion while reducing the independently governed places where it can be challenged. Research can test this by asking whether the range of conventions accepted in a literature narrows as AI-assisted writing spreads.

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Miranda Fricker's notion of hermeneutical injustice, the disadvantage suffered when a group lacks the shared interpretive resources to make its experience intelligible, identifies the harm this could produce (Fricker 2007). Kay, Kasirzadeh and Mohamed (2024) have extended Fricker's account to generative systems, identifying forms of 'generative algorithmic epistemic injustice', among them hermeneutical access injustice, illustrated by a model reporting different casualty figures depending on the language of the query, and warning of the risk of injustice when dominant groups alone set the rules of value alignment. The contribution here is not to relabel recognition displacement as epistemic injustice. It is to identify an institutional mechanism through which such injustice could arise, on which an account concerned chiefly with design and participation does not focus: if the interpretive resources of a public are increasingly shaped by systems answerable to no forum, then what can be made intelligible is settled upstream by parties whom those it disadvantages cannot reach.

6. Who made the commons?

6.1 The indexer multiplied

Nueno's history is told through its workers, librarians, documentalists, abstracters and engineers, many of them women whose contributions were minimised (Nueno 2023, pp 15, 17), and draws on Harry Braverman's account of the degradation of work (Braverman 1974). Citation indexing made authors index their own work through their references; PageRank made every web author an indexer through links; engagement ranking made every user an indexer through clicks. That the indexer had to become omnipresent in order to disappear describes a transfer of labour from a paid profession to an unpaid public (Nueno 2023, p 200), Tiziana Terranova's 'free labour' (Terranova 2000). Generative AI is trained on the textual traces of what the multiplied indexers did: not only texts but the links, citations, ratings, corrections and answers that survive as text. It learns from the record of these practices, not from the practices themselves, and does not inherit the procedures through which some of them were contested; whether a deployment retains them depends on the institutions around it.

6.2 Two grounds for a claim

Version 1 offered one ground for obligations to the commons: where AI outputs are produced with public money, openness conditions should attach (Kahl 2025a). A second ground is *provenance of contribution*. Benefit from distributed contribution cannot by itself ground it, since scholars, teachers and libraries all benefit from vast bodies of unattributed work. Its trigger is *material interference with an existing practice of traceable attribution by a service that substitutes for it*: a provider substitutes for a reference practice while materially reducing the traceability that practice provided for the function at issue. Scholarly textbooks and reviews have always reduced direct reading of the works they summarise, but they commonly preserve structured attribution through their references, if not uniformly or comprehensively; nor is the trigger the dissolution of training provenance as such,

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since a generated answer that reliably cites its supporting sources can preserve the practice. The ISI case illustrates the first ground and the argument about recognition, not this one, since the index preserved attribution exceptionally well; generative answering is the case for the provenance ground. The ground supports records of the corpus and of the attribution practices it drew on, attribution where outputs reproduce or closely derive from identifiable sources, and standing for contributors. It confers no general entitlement on every contributor to be recognised in outputs, and it is not a labour theory of property: a citation acknowledges a debt without creating a claim to payment, and converting the commons into a register of entitlements would itself be enclosure.

6.3 Depletion, net value and a missing premise

The strongest objection is that Ostrom's principles were developed for resources that use can deplete, whereas training on a text does not consume it. The objection would succeed if the commons were only a *stock* of texts. But it also has a *flow*: the continuing production, correction and renewal of the archive, on which its future value depends. Suppression threatens the flow, and there is evidence that generative AI draws contribution away. Within six months of ChatGPT's release, activity on Stack Overflow, the largest public question-and-answer site for programmers, fell by at least a quarter relative to comparable forums where the tool was less available or less capable (del Rio-Chanona, Laurentsyeva and Wachs 2024). Questions did not stop being asked; they were asked privately, of a model, and the exchange did not enter the public record.

Whether this is suppression in the sense of section 3.7 is a further question. Public value is plausible; the absence of an adequate public alternative and the attribution of the loss to substitution are not shown, since exchange may have moved elsewhere. Nor is it shown that the forum's public function was contestably governed by its users: its governance combines community voting, elected moderators and rules on duplicate questions with decisions of the company that owns it, and the question turns on whether standards of content and correction were set by moderators the community elected or by rules it made, and whether users had effective recourse against the owner. The test can be stated; its answer here is not asserted. A fall in posting is also not a fall in valuable knowledge, and users gained faster answers and, for people without libraries or English, new access. What the evidence establishes is a mechanism: a private answer can satisfy the user while failing to produce the public good a public exchange would have created, with the cost falling on future users and future models. That is the structure of an externality.

Neither benefit nor dependence alone creates an obligation. The argument needs a premise:

> *An actor whose activity makes a foreseeable and material causal contribution to a demonstrated net impairment of the continued reproduction of a shared epistemic infrastructure has a responsibility to help internalise those costs where they cannot adequately be addressed by voluntary coordination.*

Benefit from the infrastructure strengthens the responsibility but is not a condition of it; the significance of benefiting from prior contributions is the provenance ground of section 6.2.

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The externality lies in substitution, not in where the training material came from: a provider trained entirely on licensed material could still draw contribution away. The obligation arises only where an assessment shows a material causal contribution to impairment of *public epistemic capacity*, the public questioning, correction, persistence and reuse of knowledge for future users as well as present ones, not from reduced participation in a particular institution. Two difficulties follow, both of which the companion paper takes up (Kahl 2026f). The first is joint causation: where several providers together cause an impairment, each one's contribution may be immaterial on a but-for test, so that a provider-by-provider test finds no one responsible for a harm all of them caused. The second is proof. Net impairment has several dimensions, among them correction rates, the persistence of accessible solutions, the coverage of rare problems, the diversity of contributors and the ability of newcomers to acquire expertise. They do not add up to a single index, and none can be inferred from a fall in activity alone. 'Net' here means net across the dimensions of the public function itself, not net of the benefits the substituting service gives its own users; those benefits bear on what a proportionate remedy would be, not on whether the function has been impaired. The externality does not depend on the enclosure argument for its significance, though it bears on what a remedy must restore where suppression has occurred.

Any obligation that follows attaches to public functions, not to particular institutions: if a forum declines because its users obtain better answers elsewhere, the question is whether public, correctable exchange survives somewhere. Ostrom's design principles are suggestive here, since benefits from a commons should be proportionate to contributions to sustaining it (Ostrom 1990, pp 90–92; Cox, Arnold and Villamayor Tomás 2010); but that principle governs appropriators within a governed system, so the obligation extends it to external beneficiaries rather than deriving from it. It is an obligation to maintain the commons, not to pay rent for it.

6.4 What the precedent teaches

Nueno ends the ISI story with an alternative history: indexes that might have represented the variety of research rather than counting citations, which could only have mattered 'in a world where said values were deemed relevant' (Nueno 2023, p 209). The inference this paper draws is its own: a public citation index built on the same statistical principles would probably have produced many of the same effects, because the effects followed from how the index knew rather than from who held it. The point applies to Nueno's proposal for AI. Public models would be valuable counterweights to concentration, but they change who holds a system, not who settles its standards; they guard against enclosure only when their own standards are contestable.

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7. Conclusion

7.1 What the paper establishes, and what remains conditional

The paper's results are of different kinds, established to different degrees.

Level	Proposition	Status
Conceptual	Unanswerable intermediation that materially displaces a community's capacity to sustain or contest its epistemic practices is enclosure	Defined and argued (Chapter 3)
Historical	Citation indexing brought external standards into research evaluation	Supported (section 5.1)
Institutional	Such standards sometimes became presumptively governing	Plausible; stronger case-specific evidence needed (section 5.1)
Contemporary	Generative answering substitutes for activity in public forums	Supported for the setting studied (section 6.3)
Consequential	Substitution causes net impairment of public epistemic capacity	Not established (section 6.3)
Systemic	AI is producing a third enclosure movement across knowledge institutions	A thesis requiring further empirical work

The conceptual result is the definition and the distinctions that make it discriminating: origin from justification and authority; institutional answerability from ecological contestability; influence from governance; a provider's unanswerability from an institution's control over whether to use its standards; depletion from enclosure; and the grounds of duties from the conditions of enclosure. The concept is technology-neutral. AI makes it possible for its conditions to operate across general-purpose knowledge infrastructure at a scale and degree of standardisation earlier intermediaries lacked, and how far they have done so is the question the table leaves open. Enclosure is a particular failure of epistemic governance, not the general ground of duties of answerability. What the concept identifies that its components do not is a second-order loss: the displacement of the capacities through which a community could make, contest or replace the standards that govern it.

The theory also yields a testable prediction for the case it starts from. Transfer of evaluative authority to citation metrics should be detectable where research-assessment procedures create presumptions in favour of the indicators, where departures from them require additional justification, where adverse decisions track the indicators independently of other evidence, and where those assessed cannot obtain effective reconsideration of the metrics' governing role. That is a more discriminating programme than documenting how often the indicators are mentioned.

The systemic thesis can also be weakened by evidence, without touching the definition. It would be weakened if the spread of AI answering were accompanied by sustained or increasing diversity of independently governed scholarly venues; by institutions able to reject AI-derived standards without penalty; by the migration of public correction and accumulation of knowledge into new, contestably governed infrastructure; by excluded communities retaining or gaining the capacity to set standards; or by the restoration of public epistemic functions despite declining activity on incumbent platforms. Observations of this kind would challenge the application of the concept to contemporary AI, not the concept itself, and the paper's claim is open to them.

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7.2 Limitations

This version engages Nueno's book only through reports. The evidence of displacement comes from one kind of forum and one period, and should be extended to reference works, scholarly communication and libraries, with attention to net value. The position of the Global South, whose languages and knowledge practices are least represented in training data, deserves separate treatment. And the boundary between suppression and structural recognition displacement is where the theory is most open: the paper counts a field's established capacity to sustain the institutions through which it governs a practice as part of its capacity to sustain its practices, but not the loss of influence by rival institutions that survive, and whether the attributive commons owed more of its self-government to that ecology than to rights against particular gatekeepers deserves study of its own. Finally, the two mechanisms are unequally testable. For transfer, section 7.1 offers concrete indicators: presumptions in favour of indicators, burdens of justification for departing from them, decisions that track them independently of other evidence, and the absence of effective reconsideration. For suppression, the paper offers dimensions of net impairment but no comparably specific research design, because the counterfactual is harder to construct; designing one is a priority for future work. Such a design would need to establish, for a specified practice, the public function and its governance before substitution; the counterfactual trajectory of that function; changes in production, correction, accessibility and persistence; the presence or absence of functionally adequate public replacements; the attribution of the change to AI substitution rather than to other causes; and the community's dependence on unanswerable intermediaries for the immediate service.

7.3 Coda

In 1964 Taube objected that no amount of quantitative manipulation relieves anyone of the responsibility for judgement. Nueno ends his final chapter with a similar thought: however many layers conceal it, indexing 'always ends up falling back on human shoulders' (Nueno 2023, p 202). The third enclosure movement makes that fact harder to see. The companion paper asks what it would take to make those on whose shoulders judgement falls answer for it.

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Revision History

Rev	Description	Epistemic Impact	DOI	Date
0.9	LLM dissertation submitted to the University of Reading: *The Third Enclosure Movement: Rethinking AI Regulation through a Taxonomy of Restricted Knowledge in the UK, EU, US–and China*. Introduces the Third Enclosure Movement and the Taxonomy of Restricted Knowledge (legal, technical, epistemic, state-enforced); grounds them in Rawls, Marx, Ostrom and Foucault; compares the UK, EU, US and China; applies the taxonomy to healthcare, agritech and consumer AI; and argues for partial commons reforms (data trusts, audit duties, competition tools, 'Commons AI' pilots) rather than the abolition of intellectual property.	None (baseline release).	None assigned.	2025-08-24
1.0	Public release of the dissertation as a working paper (Lex et Ratio Ltd; CC BY-NC-ND 4.0), with the jurisdictions dropped from the subtitle. Adds a preface, abstract and keywords. Structure, research question, taxonomy, comparison and conclusions unchanged. Expands the doctrinal and scholarly support, notably on the limits of GDPR transparency (Koops; European Data Protection Board guidance; *Nowak*), the anticommons (Heller and Eisenberg), the US Copyright Office's reports on AI and the unsettled status of training data, US consumer-protection enforcement, China's state-led model (Zhang; Evans on embedded autonomy), knowledge-commons scholarship (Frischmann, Madison and Strandburg), the enforceability of open licences (*Jacobsen v Katzer*) and the use of TRIPS flexibilities (Baker; *Natco v Bayer*).	Low. The argument and conclusions are unchanged. The additions are supporting authorities and illustrative points, such as the anticommons risk and China's 'antitrust exceptionalism', which strengthen existing claims (chiefly that transparency duties are weak in practice and that partial commons reforms are legally feasible) without altering them.	10.5281/zenodo.17542063	2025-11-06
2.0	Full rewrite as philosophy of AI, with a new subtitle; released in the Lex et Ratio Working Papers series under CC BY 4.0. The v1 taxonomy of access is retained. The theoretical frameworks, sector case studies and comparative legal analysis are removed; the legal analysis and the remedies, including v1's partial-commons reforms, move to the companion paper *The Answerable Commons* (LXR-2026-AIGOV-ANSCOM-WP). Adds a test of effective epistemic answerability, a definition of enclosure as the displacement, by transfer or suppression, of a community's capacity to sustain and contest its epistemic practices, and a history of citation indexing as a precursor.	High. The central claim moves from restricted access to the governance of epistemic standards, and becomes more conditional: unanswerability alone is not enclosure, and the evidence does not yet establish that a third enclosure movement is occurring.	10.5281/zenodo.23643116	2026-10-25
2.1	Adds a note in §2.4 acknowledging Sonia Katyal and Christopher Morten's paper of the same title (Yale Law School, March 2026), distinguishing the enclosure of access it describes from this paper's enclosure of governance, and recording the author's earlier use of the term in the LLM dissertation (August 2025) and Version 1.0 (November 2025).	None. The argument and conclusions are unchanged.	10.5281/zenodo.23658510	2026-10-26
2.2	Revises the note in §2.4 to record the earlier, unrelated use of the phrase 'third enclosure movement' by Michael Madison (2005), for the claiming of snow-cleared parking spaces, and adds Madison (2005) to the references.	None. The argument and conclusions are unchanged.	10.5281/zenodo.23661109	2026-10-26

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